Graph Theory

Sanaa Jeehan

Week 1 — Lecture 1

Degree Sequences, Graphical Sequences & Havel–Hakimi

From vertex degrees to the question: “Can this sequence represent a simple graph?”

Learning Outcomes

By the end of this lecture, students should be able to:

- Use the Handshaking Lemma to infer structural properties of a graph.
- Write and interpret the degree sequence of a graph.
- Distinguish between an integer sequence and a graphical sequence.
- Apply necessary conditions to reject impossible degree sequences.
- Apply the Havel–Hakimi algorithm to test graphicality.
- Construct a simple graph realizing a valid degree sequence.

Prerequisite Check

A short bridge from Discrete Structures

For a simple graph G = (V,E):

Order: n = |V|

Size: m = |E|

Degree deg(v): number of edges incident with vertex v

Question: If we know only the degrees of all vertices, what can we infer about the graph?

Handshaking Lemma

$$\Sigma \deg(v) = 2|E|$$

Every edge contributes 1 to the degree of each of its two endpoints.

Therefore, the sum of all vertex degrees must be even.

Immediate consequence: every graph contains an even number of odd-degree vertices.

Quick Example: Use Degrees Without Drawing

A graph has degree sequence (4, 3, 3, 2, 2, 2).

Find the number of edges.

$$4 + 3 + 3 + 2 + 2 + 2 = 16$$

$$|E| = 16 / 2 = 8$$

The degree sequence can reveal global information even when the graph is not shown.

Degree Sequence

The degree sequence of a graph is the list of vertex degrees, usually arranged in non-increasing order.

$$(d_1, d_2, \ldots, d_n), \quad d_1 \geq d_2 \geq \ldots \geq d_n$$

Example: degrees {2, 4, 1, 2, 3} are written as (3, 4, 2, 2, 1).

Different graphs may have the same degree sequence.

Degree Sequence and Isomorphism

If two graphs are isomorphic, they must have the same degree sequence.

But the converse is not true:

- Two graphs may have identical degree sequences and still be non-isomorphic.
- Degree sequence is therefore an isomorphism invariant, but not a complete isomorphism test.

Isomorphic ⇒ Same degree sequence

Same degree sequence ⇏ Isomorphic

What Is a Graphical Sequence?

A sequence of non-negative integers is graphical if it is the degree sequence of some simple graph.

Example question:

Does (3, 3, 2, 2, 2) correspond to a simple graph?

Not every sequence of integers is graphical.

For a sequence (d_1, d_2, …, d_n) to be graphical:

Every degree must satisfy $0 \leq d_i \leq n - 1$.

The sum of the degrees must be even.

These checks can prove that a sequence is NOT graphical.

However, passing both checks does not guarantee graphicality.

Why Necessary Conditions Are Not Enough

Consider the sequence (3, 3, 3, 1).

- $n = 4$, so every degree is at most $n - 1 = 3$.
- Degree sum = 10, which is even.

Yet no simple graph realizes this sequence.

We need a stronger systematic test.

Havel–Hakimi Theorem

Start with a non-increasing sequence:

$(d_1, d_2, \ldots, d_n)$

Remove the largest degree d_1.

Subtract 1 from the next d_1 largest entries.

Reorder the remaining sequence in non-increasing order.

Repeat.

If the process reaches all zeros, the original sequence is graphical.

If a negative value occurs or the reduction becomes impossible, it is not graphical.

Havel–Hakimi: Graphical Example

Test: (3, 3, 2, 2, 2)

(3, 3, 2, 2, 2) → remove 3

(2, 1, 1, 2) → reorder → (2, 2, 1, 1)

(2, 2, 1, 1) → remove 2

(1, 0, 1) → reorder → (1, 1, 0)

(1, 1, 0) → remove 1

(0, 0) → all zeros

Conclusion: the sequence is graphical.

Havel–Hakimi: Non-Graphical Example

Test: (3, 3, 3, 1)

(3, 3, 3, 1) → remove 3

(2, 2, 0)

(2, 2, 0) → remove 2

(1, −1)

A negative entry appears.

Conclusion: the sequence is not graphical.

From a Degree Sequence to a Graph

Havel–Hakimi does more than test graphicality: it suggests how to construct a realization.

For the sequence (3, 3, 2, 2, 2):

- Choose a vertex of degree 3 and connect it to three vertices with the largest remaining degrees.
- Reduce those required degrees by 1.
- Repeat on the residual sequence.

Important: a graphical sequence can have more than one non-isomorphic realization.

Class Activity

For each sequence, decide whether it is graphical. Justify your answer.

A. (4, 3, 3, 2, 2)

B. (4, 4, 1, 1, 0)

C. (3, 3, 2, 2, 2, 2)

D. (5, 4, 3, 2, 1, 1)

For every graphical sequence, draw one realization.

Conceptual Questions

Can two non-isomorphic graphs have the same degree sequence?

Can an even degree sum guarantee that a sequence is graphical?

Can a simple graph on n vertices contain a vertex of degree n?

Can a graph have exactly one odd-degree vertex?

If all vertices have degree 2, what structural forms can the graph take?

Lecture Summary

Degree sequences provide a compact structural description of a graph.

The Handshaking Lemma gives immediate necessary constraints.

A graphical sequence is one that can be realized by a simple graph.

Simple necessary checks are useful but not sufficient.

Havel–Hakimi provides a constructive test for graphicality.

Degree sequences help with isomorphism checks, but do not uniquely identify a graph.

Practice Task

1. Determine whether (4, 4, 3, 3, 2, 2) is graphical using Havel–Hakimi.
2. If graphical, construct one realization.
3. Find the number of edges directly from the degree sequence.
4. Explain why two graphs with this degree sequence need not be isomorphic.

Next lecture: Graph Operations and Structural Transformations.